

Instructions:

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. Why do we see lightning before we hear thunder? What does this show about the speed of light and sound?
(A) Sound is heavier than light
(B) Sound changes into heat in air.
(C) Eyes are in front of ears
(D) Light travels faster than sound.
2. Why do we need to save non-renewable energy sources?
(A) Because they are unlimited.
(B) Because they will never run out.
(C) Because they take millions of years to form and are limited.
(D) Because they do not cause pollution.
3. Which of these is a way to conserve energy at home?
(A) Leaving the AC on with open windows
(B) Switching off lights when leaving a room
(C) Keeping the TV on standby all day
(D) Using incandescent bulbs instead of LEDs
4. Why do tires on a car have deep patterns called "treads" on them, especially during the rainy season?
(A) To make the car look more aerodynamic.
(B) To increase friction by channeling water away and maintaining contact with the road.
(C) To reduce the gravitational pull on the car.
(D) To make the car move faster by reducing surface area.



5. During a cricket match in Australia in late December, the players are wearing light summer clothes and sunblock. At the same time, children in England are wearing heavy coats and playing in the snow.

What is the primary reason for this difference?

- (A) England is farther from the Sun
 - (B) Sun produces less heat
 - (C) Earth's tilt makes Southern Hemisphere face the Sun
 - (D) Volcanoes keep Australia warm
6. **Assertion (A):** We should include roughage in our daily diet.
Reason (R): Roughage provides the body with a large amount of energy for physical activity.
- (A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.
 - (B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.
 - (C) Assertion (A) is true, but Reason (R) is false.
 - (D) Assertion (A) is false, but Reason (R) is true
7. Imagine Earth's rotation suddenly slowed down so that one full turn took 48 hours instead of 24. **How would this affect your daily life?**
- (A) Year becomes longer
 - (B) Day and night last 24 hours each
 - (C) Sun stops shining
 - (D) Gravity disappears
8. An animal has a body covered with scales, breathes through its lungs, and spends most of its time in the desert. **Which of the following is it most likely to be?**
- (A) A Fish
 - (B) A Frog
 - (C) A Lizard
 - (D) A Penguin
9. The given plant can be used to make ____



- (A) Cotton shirt
- (B) Nylon shirt
- (C) Raincoat
- (D) Silk saree

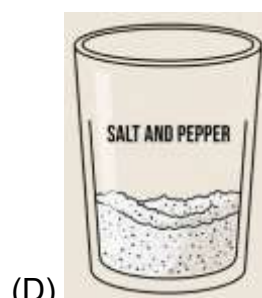
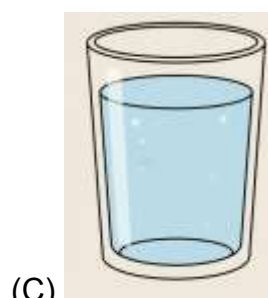
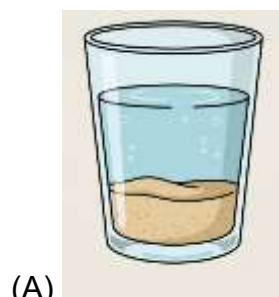
10. In nature, matter can exist in different states such as solid, liquid, and gas. When a solid is heated, it changes into a liquid, and when a liquid is heated further, it changes into a gas. This process is called interconversion of states of matter.

Which among the following represents the correct sequence of changes when ice is heated continuously?

- (A) Solid → Gas → Liquid
- (B) Solid → Liquid → Gas
- (C) Liquid → Solid → Gas
- (D) Gas → Liquid → Solid



11. Which of the following is solid-solid heterogenous mixture?



12. **Assertion:** Threshing helps in separating edible grains from the harvested crop before winnowing is carried out.

Reason: During threshing, the stalks of the crop are crushed or beaten so that the grains are released from the ears of the plant.

- (A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.
- (B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.
- (C) The assertion is true, but the reason is false.
- (D) The assertion is false, but the reason is true.

13. In a food chain: **Grass** → **Grasshopper** → **Frog** → **Snake**.



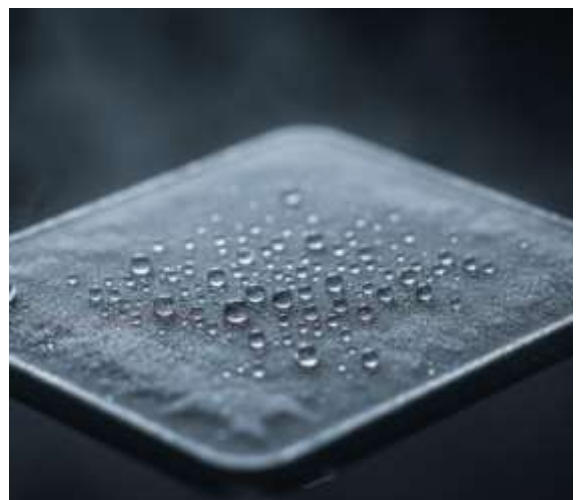
If all the frogs are removed from this area, what will happen?

- (A) The number of grasshoppers will increase.
- (B) The number of snakes will increase.
- (C) The grass will grow faster.
- (D) There will be no change.

14. A metal plate is kept in a freezer for some time and then taken out into a warm classroom. After a few minutes, tiny water droplets are seen forming on its surface.

What is the process happening here?

- (A) Evaporation
(B) Condensation
(C) Sublimation
(D) Freezing



15. Which of the following changes in the water cycle is responsible for the formation of clouds?

- (A) Evaporation
- (B) Condensation
- (C) Precipitation
- (D) Collection



16. A group of students collects salty water from a seashore for an experiment. Their teacher asks them to obtain pure drinking water from this sample in such a way that salt and other dissolved impurities are completely removed.

Which method of water purification should they use?

- (A) Boiling
- (B) Filtration
- (C) Chlorination
- (D) Distillation

17. Look at the classification below:

Group X: Cow, Goat, Deer

Group Y: Lion, Tiger, Wolf Which animal can be placed in Group X?

- (A) Hyena
- (B) Zebra
- (C) Leopard
- (D) Polar Bear

18. Select the **incorrect statement** related to road safety rules

- (A) Pedestrians should use zebra crossings and look left, right, and left again before crossing the road.
- (B) Cyclists should wear helmets and follow traffic signals while riding.
- (C) Passengers should always lean out of moving vehicles to check traffic conditions.
- (D) People should walk on the footpath whenever it is available.

19. Unscramble the letters in the given options to find out the name of an item used in a first aid box to cover cuts and wounds.



- (A) EGADNAB
- (B) SROCSISS
- (C) CITESPETNA
- (D) EVSEIDAEPT

20. In the Water Cycle, the process of 'Precipitation' refers to:

- (A) Water turning into vapor
- (B) Water falling as rain or snow
- (C) Plants releasing water
- (D) Clouds forming from vapor

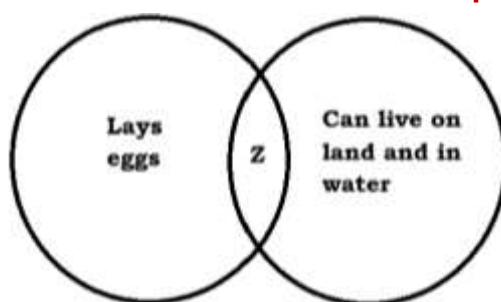
21. Identify the organ: **"I am a muscular bag. I churn food and mix it with digestive juices."**

- (A) Heart (B) Stomach (C) Lungs (D) Liver

22. A person's diet consists only of polished rice and sugar. After some time, they might suffer from a deficiency disease because their diet lacks:

- (A) Energy (B) Vitamins and Minerals
(C) Carbohydrates (D) Taste

23. Study the Venn diagram below. **Which animal best represents 'Z'?**



- (A) Whale (B) Frog (C) Fish (D) Bat

24. When you run a 100-meter race, your heart beats faster and you breathe more quickly. **Why does this happen?**

- (A) Because your heart is scared of the race.
(B) Your muscles need more oxygen and energy to work harder.
(C) To help you cool down your body temperature instantly.
(D) To push the food out of your stomach faster.

25. In a village, people started using plastic bags and throwing them into the local drains. During the monsoon, the village flooded. **What is the most likely reason?**

- (A) Plastic bags turned into water.
(B) Plastic bags blocked the flow of water in the drains.
(C) Plastic bags attracted more rain.
(D) The soil absorbed too much plastic.

26. The temperature on the day side of Mercury is 430°C, while the night side drops to – 180°C. **What is the total temperature "swing" (difference) between day and night on Mercury?**

- (A) 250°C (B) 430°C (C) 610°C (D) 300°C

27. A satellite orbits the Earth 15 times in exactly 24 hours. How long does it take for this satellite to complete one single orbit?

- (A) 96 minutes (B) 60 minutes (C) 120 minutes (D) 45 minutes

28. Look at the following items shown below.



(i)



(ii)



(iii)



(iv)

Which of them can be worn during winter?

(A) (i),(ii)

(B) (ii), (iii), (iv)

(C) (ii), (iii)

(D) (iv), (iii)

29. **Assertion (A):** We should not sleep under trees at night.

Reason (R): During the night, plants take in carbon dioxide and release oxygen.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

(B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

30. When you accidentally touch a hot object, your hand moves away instantly. **Which part of the Nervous System is responsible for this "Reflex Action"?**

(A) Brain

(B) Heart

(C) Spinal Cord

(D) Stomach

31. Observe the foot structure of an unknown bird: it has two toes pointing forward and two toes pointing backward with sharp claws. Based on this, **what is the most likely lifestyle of this bird?**

(A) It swims in ponds to find fish.

(B) It wades in shallow water to catch frogs.

(C) It climbs vertical tree trunks to find insects in the bark.

(D) It runs very fast on open grasslands.

32. Select the option that **correctly pairs** the layer of the atmosphere with its main feature.

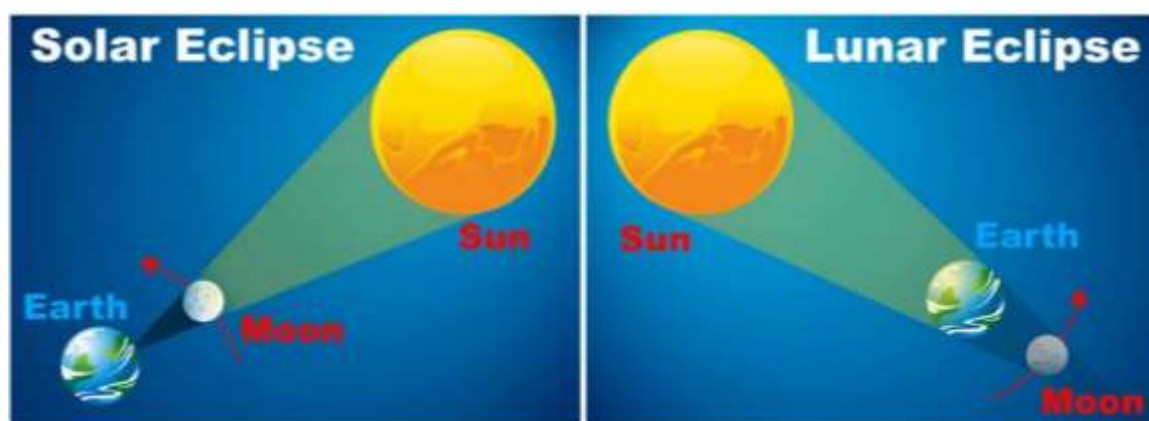
(A) Troposphere – Contains the ozone layer that protects from UV rays

(B) Stratosphere – Coldest layer where meteors burn up

(C) Mesosphere – Coldest layer where meteors burn up

(D) Exosphere – Layer where weather occurs and planes fly

33. Which of the following is a cause of **air pollution**?
- (A) Planting more trees (B) Burning garbage in open areas
- (C) Using bicycles instead of cars (D) Recycling waste materials
34. Venus is often called Earth's "sister planet" because of its size, but it is the hottest planet in the solar system, even hotter than Mercury **which is closer to the Sun. Why?**
- (A) Many volcanoes produce heat (B) Thick CO₂ atmosphere traps heat
- (C) Fast rotation causes heating (D) Hot metal planet
35. Why is a Solar Eclipse visible only from a very small area on Earth, while a Lunar Eclipse can be seen by half of the world at once?



- (A) The Moon's shadow is very small compared to the Earth's shadow.
- (B) Solar eclipses only happen at the North Pole.
- (C) The Sun is smaller than the Moon, making its shadow tiny.
- (D) Lunar eclipses only happen during the day when everyone is awake.

★ ★ ★